

Standing Carrier Identity Closure

A Same-Scope Exhaustion Theorem for Source-Distinct Physical Realizations under AASC

Amos Jay Maley
Independent Researcher

August 2026

Abstract

A source-first physical realization program may generate several constructions that reproduce the same fixed target while differing in primitive field inventory, source graph, ultraviolet continuation, or other provenance data. The mere existence of those source distinctions does not yet determine whether they are distinct *standing carrier identities* at the target scope. This paper isolates that identity question and proves a clean-room fixed-domain theorem.

Fix a target-frozen carrier comparison scope with a carrier-complete standing interface and licensed continuations. The full continuation signature of a carrier presentation is the tuple of all admitted standing-bearing witness outputs after all licensed continuations. We prove that equality of this full signature induces the unique same-scope standing carrier identity quotient compatible with stable reference, carrier descent, identity-preserving redescription, reuse consistency, bivalent standing, no same-act repair, and the absence of a second same-scope admissibility-bearing classifier. Any finer quotient retains an unwitnessed distinction and is therefore over-resolved; any coarser quotient identifies a target-witnessed difference and is therefore under-resolved. A separate activation theorem makes the substantive application burden explicit: the witness/continuation interface must be frozen independently and shown to exhaust the target's definedness, quotient-factorizing invariant, compatibility, failure, and continuation loci before carrier counting begins.

At the carrier layer, every elementary descriptive variation is anchor, tensor, or skin. Finite composite source differences decompose into anchor motion together with tensor-determined same-scope content and skin/provenance remainder. Consequently, source or provenance labels do not acquire terminal carrier-identity authority merely by being retained in a candidate registry. Added source structure either changes no standing-bearing target content and collapses at carrier resolution, changes an already admitted tensor coordinate and remains genuinely carrier-distinct, or requires a different target/interface and therefore belongs to a new certificate.

The identity and realization stages are then separated. First form the final standing carrier quotient. If it is singleton, standing-carrier identity is closed. If it is genuinely non-singleton, same-scope identity collapse is no longer available: the original fixed certificate can reach a structural pre-lock singleton only by an independently formed realized-carrier trace or by impossibility exclusion of all but one standing class. Empirical tie-breaks, imported selectors, post-hoc representative choice, and undeclared scope changes are not same-scope derivations. The result therefore supplies a fail-closed structural test for whether source multiplicity is genuine carrier multiplicity or only over-resolution of one standing carrier identity, while preserving genuinely witness-distinct alternatives when neither trace nor exclusion is available.

Contents

1	Problem and theorem posture	2
1.1	Clean-room boundary	2
1.2	Main result in compact form	2
2	Imported structural substrate	3
3	Fixed carrier scope and continuation signatures	4
4	Uniqueness of terminal carrier identity	6
5	Source multiplicity versus carrier multiplicity	7
6	No hidden carrier classifier	8
7	Same-scope carrier-distinction exhaustion	9
8	Realized-carrier route exhaustion	10
9	Singleton carrier theorem	12
10	Negative control: when collapse must not occur	13
11	Counterexample burden and hostile tests	13
12	Relation to the supplied structural corpus	14
13	Scope and nonclaims	15
14	Conclusion	16
A	Dependency map	16
B	Application protocol	17
C	Compact theorem package	18

1 Problem and theorem posture

A source-first exhaustion program correctly preserves distinctions that might matter. That preservation is essential during candidate generation: if two proposals arise from different primitive source architectures, merging them before the target comparison is complete can erase a genuine physical alternative. The converse danger appears later. A provenance distinction that was useful for candidate generation may be carried forward into a terminal identity quotient even after the target has supplied no standing-bearing witness for it. In that case the terminal quotient becomes finer than the standing structure of the fixed target.

This paper addresses only that second problem. It asks:

When do two source-distinct physical realizations constitute two standing carrier identities in one fixed target, and when are they only distinct source presentations of one target-relative carrier identity?

The proof form is AASC-native: fix the target and comparison class, define the possible loci at which a same-scope distinction can carry standing, and exhaust every route by which a source difference could survive as terminal carrier identity.

1.1 Clean-room boundary

No particular downstream carrier problem is imported as a premise. In particular, the manuscript does not assume a specific particle content, neutrino completion, Standard-Model extension, or previously computed carrier cardinality. The only imported results are fixed-domain structural constraints on admissibility, standing, stable reference, identity-preserving redescription, no-repair, and same-scope classifier uniqueness. The theorem is therefore usable as a later application constraint without being tailored to any one residue calculation.

1.2 Main result in compact form

Let $\mathcal{S} = (\mathcal{P}, \mathcal{W}, \mathcal{U}, \mathcal{T})$ be a fixed target-frozen carrier comparison scope. For each carrier presentation $x \in \mathcal{P}$, define its complete standing continuation signature

$$\Sigma_{\mathcal{S}}(x) := (w(u(x)))_{(w,u) \in \mathcal{W} \times \mathcal{U}}.$$

Define

$$x \sim_{\text{car}} y \iff \Sigma_{\mathcal{S}}(x) = \Sigma_{\mathcal{S}}(y).$$

The central theorem proves that, under the structural hypotheses stated below, $\sim_{\text{car}}$ is the unique admissible same-scope standing carrier identity relation. Therefore

$$Q_{\text{car}}(\mathcal{S}) = \mathcal{P} / \sim_{\text{car}} \cong \text{im}(\Sigma_{\mathcal{S}}).$$

Source multiplicity and standing carrier multiplicity coincide only when the source distinctions are visible in the complete standing continuation signature. Once this quotient is formed from a certified complete interface, it is final for same-scope identity: if it is non-singleton, only trace realization or impossibility elimination can produce a structural pre-lock singleton under the original target.

2 Imported structural substrate

The paper uses a deliberately small structural substrate. The following clauses are not re-derived here; they are imported from the fixed-domain AASC corpus and restated only to make every dependency explicit. The kernel and admissibility-structure results supply the governance roles, bivalence, no-repair, standing conservation, and unique-interior discipline [1, 2]. Stable reference and continuation-signature quotienting are supplied in fixed-domain form by [3]; the descriptive-carrier descent, composite effect closure, and no-refinement-beneath-tensor-content results are supplied explicitly by the Anchor–Tensor–Skin theorem [6]; and independent no-second-classifier support is supplied by [4, 5].

Definition 2.1 (Kernel-governed fixed-domain use). *A carrier comparison is kernel-governed when its same-scope use preserves the four governance roles*

$$K = \{\text{Adm}, \text{St}, \text{Ref}, \text{Irr}\}.$$

Reference fixes the target and identity being carried; standing marks eligibility for licensed continuation and reuse; irreversibility forbids post-hoc repair from counting as the same act or realization; admissibility is the gate under which these roles remain jointly enforceable.

Definition 2.2 (Stable reference). *Fix a presentation class $\mathcal{P}$, admitted target tests $\mathcal{W}$, and licensed continuations $\mathcal{U}$. Stable reference requires that an identity-bearing distinction be supported by the admitted interface: if two presentations remain indistinguishable under every admitted witness after every licensed continuation, they may not be maintained as two distinct standing states of the same fixed scope merely by retaining different labels or provenance records.*

Definition 2.3 (Bivalent standing and no repair). *Standing on the evaluated fragment is bivalent. A standing-negative act or realization cannot be made standing-positive as the same act or realization by downstream reinterpretation, representative choice, parameter preference, or repair. A legitimate change of identity or target requires explicit reset and fresh gating.*

Definition 2.4 (No second same-scope authorizer). *No auxiliary predicate, ranking, selector, probability, canonical representative, provenance label, or structural score may refine the same fixed-domain standing partition while claiming independent admissibility-bearing authority. If such an object changes no licensed use, it is bookkeeping; if it changes licensed use, it functions as a second gate or a scope-changing tensor/anchor.*

Definition 2.5 (Anchor–tensor–skin carrier descent). *Fix a finite descriptive carrier and vary one component while holding the remaining fixed-domain typing data constant. Every such elementary variation has exactly one of three effects:*

1. *anchor: the variation changes target typing, definedness, boundary data, or the evaluated domain;*
2. *tensor: anchors remain fixed, but the variation changes a standing-bearing quotient consequence or licensed continuation;*
3. *skin: the variation changes neither target typing nor any standing-bearing consequence or continuation.*

No fourth same-scope standing-bearing effect channel is available. Finite joint or relational effects factor through anchor motion or tensor content; if both are fixed, the remaining variation is skin/bookkeeping [6].

Remark 2.6 (Why singleton governance is not imported as carrier singleton). *The necessity of singleton governance is relevant only as a block on arbitration or second-gate authority. It is not used to infer that a physical residue must itself be singleton. One governance regime may coherently diagnose a non-singleton residue. Carrier singletonhood must therefore be established at the carrier identity quotient or by a lawful realization route, not by a level jump from governance cardinality.*

3 Fixed carrier scope and continuation signatures

Definition 3.1 (Target-frozen carrier comparison scope). *A target-frozen carrier comparison scope is a tuple*

$$\mathcal{S} = (\mathcal{P}, \mathcal{W}, \mathcal{U}, \mathcal{T})$$

with the following roles.

- $\mathcal{P}$ is a class of source-generated carrier presentations admitted for comparison.
- $\mathcal{T}$ is the fixed target certificate: domain, target work, standing roles, lawful equivalence conventions, and target-level scope.
- $\mathcal{W}$ is the independently frozen family of target-admitted structural witnesses. A witness need not be an empirical detector readout; it may be any target-authorized invariant, boundary record, compatibility relation, failure condition, or structural consequence that carries standing.
- $\mathcal{U}$ is the monoid of licensed same-scope continuations or updates under which target identity is required to remain stable.

Definition 3.2 (Witness completeness). *The witness family $\mathcal{W}$ is target-complete when every standing-bearing same-scope difference between two carrier presentations is exposed by at least one admitted witness after some licensed continuation. Formally, whenever a same-scope difference changes target-relative physical standing, admissible continuation, compatibility, failure status, or identity-preserving reuse, there exist $w \in \mathcal{W}$ and $u \in \mathcal{U}$ for which the witness outputs differ.*

Definition 3.3 (Carrier-interface activation certificate). *A target-frozen scope carries a carrier-interface activation certificate when, before the terminal carrier endpoint is inspected, it records:*

1. *the complete target/definedness and boundary interface;*
2. *the complete family of target-admitted quotient-factorizing invariants, compatibility relations, failure conditions, and other standing-bearing consequence functionals;*
3. *the licensed same-scope continuation monoid $\mathcal{U}$; and*
4. *evidence that the preceding families are formed from the target certificate and upstream standing rules rather than from the desired carrier count.*

The witness family $\mathcal{W}$ is then the family of evaluation maps associated with those standing-bearing consequence functionals, together with the target-admitted boundary and failure witnesses already present in the certificate.

Theorem 3.4 (Carrier-interface activation theorem). *If a fixed target satisfies the carrier-interface activation certificate and the imported carrier-descent and no-refinement theorems apply, then the induced witness family $\mathcal{W}$ is target-complete in the sense of the preceding definition.*

Proof. By carrier descent, an elementary descriptive variation either changes definedness/target typing, changes tensor-determined standing content, or changes neither. Anchor changes are outside the fixed same-scope comparison. With anchors fixed, the imported no-refinement theorem states that coherent retained-trajectory behavior is tensor-determined and that no stricter admissibility-preserving refinement survives beneath fixed tensor content [6]. By the activation certificate, every target-admitted tensor consequence, compatibility, failure condition, and licensed continuation is represented in the frozen interface. Hence every standing-bearing same-scope difference changes at least one induced witness after some licensed continuation. That is target completeness. $\square$

Remark 3.5 (Fail-closed application gate). *The activation certificate is an application gate, not an endpoint selector. It must be discharged before the terminal carrier count is inspected. If an application cannot establish it, SCIC returns interface incomplete / not ready for carrier-identity closure; it does not license a singleton quotient.*

Definition 3.6 (Complete standing continuation signature). *For $x \in \mathcal{P}$, define*

$$\Sigma_{\mathcal{S}}(x) := (w(u(x)))_{(w,u) \in \mathcal{W} \times \mathcal{U}}.$$

The codomain is the product of the witness-output spaces over all admitted witness/continuation pairs. The signature records only target-authorized standing-bearing content. Candidate provenance may be retained elsewhere without automatically entering $\Sigma_{\mathcal{S}}$.

Definition 3.7 (Standing carrier equivalence). *For $x, y \in \mathcal{P}$, define*

$$x \sim_{\text{car}} y \iff \Sigma_{\mathcal{S}}(x) = \Sigma_{\mathcal{S}}(y).$$

The corresponding standing carrier quotient is

$$Q_{\text{car}}(\mathcal{S}) := \mathcal{P} / \sim_{\text{car}}.$$

Lemma 3.8 (Continuation congruence). *The relation $\sim_{\text{car}}$ is an equivalence relation and is stable under every licensed continuation $u_0 \in \mathcal{U}$.*

Proof. Equality of complete signatures is reflexive, symmetric, and transitive. Suppose $x \sim_{\text{car}} y$. For any $u_0 \in \mathcal{U}$, any $w \in \mathcal{W}$, and any $u \in \mathcal{U}$, closure of $\mathcal{U}$ under composition gives $u \circ u_0 \in \mathcal{U}$. Hence

$$w(u(u_0(x))) = w((u \circ u_0)(x)) = w((u \circ u_0)(y)) = w(u(u_0(y))).$$

Thus $u_0(x) \sim_{\text{car}} u_0(y)$. $\square$

Theorem 3.9 (Intrinsic carrier detectability). *If $[x] \neq [y]$ in $Q_{\text{car}}(\mathcal{S})$, then there exist an admitted witness $w \in \mathcal{W}$ and a licensed continuation $u \in \mathcal{U}$ such that*

$$w(u(x)) \neq w(u(y)).$$

Proof. Distinct quotient classes mean $\Sigma_{\mathcal{S}}(x) \neq \Sigma_{\mathcal{S}}(y)$. Inequality in the product signature implies inequality in at least one coordinate, which is exactly a pair (w, u) . $\square$

4 Uniqueness of terminal carrier identity

The preceding quotient is not introduced merely as a convenient equivalence. Stable reference makes it the only admissible same-scope identity relation compatible with witness soundness and witness completeness.

Definition 4.1 (Candidate terminal carrier identity relation). *A candidate terminal carrier identity relation is an equivalence relation $\equiv$ on $\mathcal{P}$ proposed to determine which source-generated presentations count as the same standing carrier at the fixed target.*

Theorem 4.2 (Stable-reference constraints on carrier identity). *Let $\equiv$ be a candidate terminal carrier identity relation on a target-frozen, witness-complete scope. If $\equiv$ is to carry same-scope standing identity under stable reference, it must satisfy all four conditions below:*

1. witness soundness: $x \equiv y$ implies agreement of every admitted continuation witness;
2. witness completeness: agreement of every admitted continuation witness implies $x \equiv y$;
3. continuation stability: $x \equiv y \Rightarrow u(x) \equiv u(y)$ for every licensed continuation;
4. reuse consistency: identity cannot depend on which lawful representative or source label is used during same-scope reuse.

Proof. If a candidate relation identifies presentations separated by an admitted continuation witness, the same standing identity would carry two incompatible target-authorized consequences, violating reference stability and determinate reuse. This proves witness soundness. If it separates presentations indistinguishable by every admitted continuation witness, the distinction has no target-authorized standing effect; retaining it as identity-bearing therefore requires an unwitnessed discriminator or source-label authorizer, forbidden by stable reference and the no-second-classifier closure. This proves witness completeness. Continuation stability is required because an identity relation that ceases to preserve identity under a licensed same-scope continuation does not support standing-bearing reuse. Reuse consistency follows for the same reason: representative-dependent identity would make future standing depend on presentation rather than the fixed target object. Thus all four conditions are forced by the imported fixed-domain structure rather than stipulated as a desired quotient. $\square$

Definition 4.3 (Admissible carrier identity relation). *A candidate terminal carrier identity relation is admissible when it satisfies the four structurally forced conditions of the preceding theorem.*

Theorem 4.4 (Unique standing carrier identity quotient). *Let $\mathcal{S}$ be target-frozen and witness-complete. Every admissible terminal carrier identity relation $\equiv$ coincides with $\sim_{\text{car}}$. Hence $Q_{\text{car}}(\mathcal{S})$ is the unique same-scope standing carrier identity quotient.*

Proof. Assume $x \equiv y$. Witness soundness gives equality of all $w(u(\cdot))$ coordinates, so $\Sigma_{\mathcal{S}}(x) = \Sigma_{\mathcal{S}}(y)$, hence $x \sim_{\text{car}} y$. Conversely, if $x \sim_{\text{car}} y$, all admitted continuation witnesses agree. Witness completeness then gives $x \equiv y$. Thus $\equiv$ and $\sim_{\text{car}}$ contain exactly the same ordered pairs. Continuation stability follows from the previous lemma; reuse consistency blocks representative-dependent refinements of this relation. $\square$

Corollary 4.5 (No alternative same-scope carrier quotient). *No alternative terminal carrier identity quotient can be both same-scope and standing-correct. A finer quotient separates presentations not separated by any admitted target witness; a coarser quotient identifies presentations that an admitted target witness distinguishes.*

Proof. A finer quotient violates witness completeness and stable reference. A coarser quotient violates witness soundness. Either way the proposed quotient is not an admissible terminal carrier identity relation. $\square$

Corollary 4.6 (Carrier quotient as signature image). *There is a canonical bijection*

$$Q_{\text{car}}(\mathcal{S}) \cong \text{im}(\Sigma_{\mathcal{S}}).$$

In particular,

$$|Q_{\text{car}}(\mathcal{S})| = |\text{im}(\Sigma_{\mathcal{S}})|.$$

Proof. The quotient identifies exactly the fibers of $\Sigma_{\mathcal{S}}$. The standard first-isomorphism argument for sets sends $[x]$ to $\Sigma_{\mathcal{S}}(x)$. $\square$

5 Source multiplicity versus carrier multiplicity

Source-first generation correctly retains source labels, primitive inventories, provenance trees, and construction routes. The next results determine when those distinctions may enter terminal carrier identity without treating a composite source change as though it were one elementary coordinate variation.

Definition 5.1 (Source distinction). *Let $s : \mathcal{P} \rightarrow \mathcal{S}_{\text{src}}$ be any source/provenance record: primitive node inventory, source graph, UV completion label, construction route, operator origin, or other generation-level descriptor. A pair $x, y \in \mathcal{P}$ is source-distinct when $s(x) \neq s(y)$.*

Definition 5.2 (Elementary source variation). *Fix a finite source-description tuple $D = (C_1, \dots, C_n)$. An elementary source variation changes one component C_i while the remaining fixed-domain typing data are held constant. Composite source differences are finite families of such elementary variations, possibly with joint or relational effects.*

Theorem 5.3 (Elementary source-difference role trichotomy). *Fix an activated target-complete carrier scope $\mathcal{S}$, and let δ_i be an elementary source variation. Exactly one target-relative role applies.*

1. **Anchor difference.** *Varying δ_i changes the target, evaluated fragment, boundary typing, or admitted witness/continuation interface. The variation is not same-scope.*
2. **Tensor difference.** *anchors remain fixed and varying δ_i changes $\Sigma_{\mathcal{S}}$. The variation is a genuine standing-bearing same-scope carrier difference.*
3. **Skin/provenance difference.** *anchors remain fixed and varying δ_i changes no coordinate of $\Sigma_{\mathcal{S}}$. The variation may remain in source provenance but carries no terminal same-scope carrier-identity authority.*

No fourth standing-bearing elementary source role exists.

Proof. This is the activated carrier-descent theorem. If definedness or target typing changes, the variation is anchor. Assume anchors remain fixed. Target completeness and the no-refinement theorem force every remaining standing-bearing effect into tensor-determined continuation content. Hence either the complete signature changes, giving tensor, or no standing-bearing effect changes, giving skin. Pairwise disjointness follows from the role definitions; a fourth effect channel is excluded by carrier descent [6]. $\square$

Theorem 5.4 (Composite source-difference closure). *Let $E = \{\delta_i\}_{i \in I}$ be a finite family of elementary source variations, including any finite joint or relational source effect. Then its total target-relative effect decomposes into*

$$\begin{aligned} \text{total source effect} = & \text{anchor motion} + \text{tensor-determined same-scope content} \\ & + \text{skin/provenance remainder.} \end{aligned}$$

If anchor motion is absent, every standing-bearing composite effect factors through Σ_S ; if the signature is also fixed, the entire residual composite difference is skin/bookkeeping.

Proof. The carrier-descent composite-closure theorem states that no finite combination of descriptive elements creates a new same-scope admissibility-bearing effect channel beyond definedness/anchor work and tensor-level invariant-profile work, with all remaining content inert [6]. The activated signature contains those target-admitted tensor consequences after licensed continuation. Therefore the joint effect factors through the stated decomposition. $\square$

Corollary 5.5 (No source-label promotion). *Source distinctness alone does not imply carrier distinctness:*

$$s(x) \neq s(y) \not\Rightarrow [x] \neq [y] \text{ in } Q_{\text{car}}(\mathcal{S}).$$

A source label separates terminal carrier identities only through tensor-bearing content already present in the frozen activated target interface.

Corollary 5.6 (Added-source disposition). *Let $y = x \oplus \Delta L$ be obtained by adding a finite source package ΔL to x . Evaluate the package in the following priority order.*

1. *If some component changes target typing, definedness, boundary conditions, or the admitted interface, the extension is target/scope change.*
2. *Otherwise, if the composite package changes Σ_S , it contains genuine tensor-bearing same-scope content and $[x] \neq [y]$.*
3. *Otherwise, $\Sigma_S(x) = \Sigma_S(y)$, so $x \sim_{\text{car}} y$ and the residual added structure is terminally skin/provenance at that target resolution.*

These dispositions are disjoint under the stated priority and exhaustive for finite added source structure.

Remark 5.7 (Candidate-generation labels remain legitimate). *Nothing here requires deleting source labels from a candidate registry. Provenance remains essential for generation completeness, audit, and later target extensions. The theorem denies only an automatic promotion from candidate-generation identity to terminal standing carrier identity.*

6 No hidden carrier classifier

Definition 6.1 (Auxiliary carrier classifier). *An auxiliary carrier classifier is a map*

$$K : \mathcal{P} \rightarrow Z$$

whose values are proposed to distinguish terminal carrier identities beyond the quotient $Q_{\text{car}}(\mathcal{S})$. Let $q_{\text{car}} : \mathcal{P} \rightarrow Q_{\text{car}}(\mathcal{S})$ denote the quotient map. We call K quotient-factorizing when there exists $\bar{K} : Q_{\text{car}}(\mathcal{S}) \rightarrow Z$ with $K = \bar{K} \circ q_{\text{car}}$.

Theorem 6.2 (No second same-scope carrier classifier). *Let K be an auxiliary carrier classifier for an activated fixed target, together with its proposed carrier-identity use. Apply the following decision order. Exactly one primary status results.*

1. **Quotient-factorizing.** *If $K = \bar{K} \circ q_{\text{car}}$, then K contains no finer carrier identity than Q_{car} ; it is a renaming, report, or downstream function of already standing carrier information.*
2. **Interface-changing proposal.** *Otherwise, if the proposed standing force of K requires changing the witness/continuation interface, target typing, or boundary package, the proposal belongs to a new target/certificate rather than the original same-scope quotient.*
3. **Nonfactorizing but standing-inert.** *Otherwise the original interface is fixed. If K varies within a $\sim_{\text{car}}$ -class but changes no licensed target use, continuation, compatibility, failure status, or standing consequence, it is skin/provenance.*
4. **Nonfactorizing same-scope authority.** *In the remaining fixed-interface case, K varies within a $\sim_{\text{car}}$ -class and is asserted to change terminal carrier identity or licensed use. It is an inadmissible second same-scope standing classifier.*

No fifth same-scope carrier-authority status survives.

Proof. The first decision separates quotient-factorizing from nonfactorizing classifiers. For a nonfactorizing proposal, the second decision separates explicit target/interface change from same-scope use. With the original interface fixed, carrier descent is bivalent with respect to standing effect: the variation either changes no standing-bearing use and is skin, or is asserted to change such use and thereby refines a complete signature fiber. The latter is the forbidden second classifier. The decision order makes the statuses pairwise disjoint, while carrier descent, quotient finality, and no-second-classifier closure make them exhaustive [6, 4]. $\square$

Corollary 6.3 (No simplicity or provenance selector). *Simplicity, field count, operator dimension, naturalness, historical source label, preferred UV narrative, canonical representative, or any analogous preference cannot refine terminal carrier identity inside one $\sim_{\text{car}}$ -class. If it factors through Q_{car} , it merely reports existing standing information; if it varies inside a class without changing target use, it is skin; if it claims independent same-scope authority, it is inadmissible; and if it becomes authoritative only after an interface change, it belongs to a new target.*

7 Same-scope carrier-distinction exhaustion

We now state the clean-room constraint in a disjoint adjudication form.

Definition 7.1 (Carrier-distinction claim). *A carrier-distinction claim for a pair (x, y) asserts that the presentations represent distinct terminal standing carriers. The claim is same-scope only if target typing, boundary package, witness family, continuation monoid, and evaluated fragment remain those of the already activated scope S .*

Theorem 7.2 (Same-Scope Carrier-Distinction Exhaustion). *Every carrier-distinction claim has exactly one primary adjudication status relative to the activated target.*

1. **Target/scope change.** *The claim cannot be made standing-bearing without changing the target, boundary package, witness family, continuation family, or evaluated fragment. It is therefore not a same-scope carrier rivalry under S .*

2. **Standing-quotient collapse.** *The scope is preserved and $\Sigma_S(x) = \Sigma_S(y)$. Then $x \sim_{\text{car}} y$: there is one terminal standing carrier identity with possibly distinct source/provenance presentations.*
3. **Witnessed same-scope distinction.** *The scope is preserved and $\Sigma_S(x) \neq \Sigma_S(y)$. Then an already admitted witness/continuation coordinate separates the pair, and they are genuinely distinct standing carriers.*

No fourth semantic adjudication status exists.

Proof. First ask whether adjudicating the claimed distinction preserves the activated target interface. If not, status (1) applies. If it does, the complete signatures are well defined in the same scope and obey classical equality: either they agree or they do not. Equality gives status (2) by uniqueness of the standing carrier quotient. Inequality gives status (3), and intrinsic detectability supplies an already admitted separating witness/continuation coordinate. These alternatives are pairwise disjoint and exhaustive. $\square$

Corollary 7.3 (Illicit-refinement failure normal form). *If a same-scope claim falls under standing-quotient collapse but the two presentations are nevertheless retained as distinct terminal carriers by provenance, representative choice, ranking, weighting, repair, simplicity, or another auxiliary authorizer, that retention is not a fourth semantic carrier status. It is a failure normal form: a nonfactorizing second classifier inside one standing-signature fiber.*

Corollary 7.4 (Candidate multiplicity does not imply carrier multiplicity). *A source-first candidate universe can be non-singleton even when the terminal standing carrier quotient is singleton. Conversely, a genuinely witness-distinct target residue remains non-singleton regardless of how many source labels are quotiented away.*

Corollary 7.5 (Standing carrier cardinality formula). *For any nonempty candidate family $A \subseteq \mathcal{P}$,*

$$|A / \sim_{\text{car}}| = |\Sigma_S(A)|.$$

Thus exact terminal carrier cardinality is an exact signature-image cardinality problem, not a source-record count.

8 Realized-carrier route exhaustion

The previous theorem answers the identity question and produces the *final same-scope standing carrier quotient*. A second question remains only when that quotient itself is non-singleton: which already distinct class, if any, is structurally realized? AOT-style prediction-lock analysis supplies the correct precedent. Candidate separation is not realization, and after the final standing quotient is formed there is no further same-scope identity-collapse route.

Definition 8.1 (Target-admitted trace-record domain). *Let $\mathcal{R}_{\mathcal{T}}^{\text{tr}}$ be the pre-frozen domain of same-scope structural records that the target certificate permits to bear realized-carrier trace information. Membership in $\mathcal{R}_{\mathcal{T}}^{\text{tr}}$ and the interpretation of its records must be fixed independently of the desired carrier endpoint.*

Definition 8.2 (Realized-carrier trace). *Let $Q = A / \sim_{\text{car}}$ be the final standing carrier quotient of a live candidate family A . The fixed target may contain a preformed stable incidence predicate*

$$I_{\text{tr}} : \mathcal{R}_{\mathcal{T}}^{\text{tr}} \times Q \longrightarrow \{0, 1\}$$

whose formation is independent of the desired carrier endpoint. A realized-carrier trace is a certified record $r_{\text{real}} \in \mathcal{R}_{\mathcal{T}}^{\text{tr}}$ for which the incidence fiber is a singleton:

$$\{q \in Q : I_{\text{tr}}(r_{\text{real}}, q) = 1\} = \{q_*\}.$$

The record therefore fixes exactly one already standing carrier class without changing the target. Neither I_{tr} nor r_{real} may be constructed from post-hoc empirical preference, representative choice, or an imported selector.

Definition 8.3 (Impossibility realization). *An impossibility realization is a same-scope theorem establishing that every $q \in Q \setminus \{q_*\}$ is impossible under the fixed target certificate, leaving exactly one nonempty standing class q_* .*

Theorem 8.4 (Post-quotient singleton-realization exhaustion). *Let $Q = A / \sim_{\text{car}}$ be the final standing carrier quotient produced from an activated target-complete scope, and suppose $|Q| > 1$. Then a structural pre-lock singleton realized carrier under the original fixed certificate can arise only by one of two lawful routes:*

1. **trace realization:** *an independently formed target-admitted realized-carrier trace fixes one already standing class; or*
2. **impossibility realization:** *all rival standing classes are excluded by same-scope impossibility.*

Empirical comparison is post-lock rather than a derivation from the original certificate; an imported selector is inadmissible; and target/interface change creates a new certificate. Identity collapse is not an available post-quotient route because distinct elements of Q are already separated by the complete standing signature.

Proof. Because Q is the final quotient by complete signature equality, distinct $q_1, q_2 \in Q$ cannot be merged by any alternative same-scope carrier identity relation without violating witness soundness and quotient finality. Hence realization cannot proceed by another identity collapse. A structural singleton must therefore either positively fix one already standing class by a target-admitted trace or negatively eliminate every rival by impossibility. Any remaining determination source is external comparison, imported authority, or changed scope, yielding post-lock determination, inadmissibility, or a new certificate. The AOT route-exhaustion pattern independently confirms this staging [16]. $\square$

Corollary 8.5 (No singleton from separation alone). *Showing that final standing carrier classes are mathematically or physically different does not itself realize one of them. If $|Q| > 1$, witnessed distinction without trace or impossibility leaves the target structurally underdetermined at carrier resolution.*

Theorem 8.6 (Staged carrier-closure theorem). *Let $A \subseteq \mathcal{P}$ be a nonempty live candidate family for an activated target-complete scope. Form*

$$Q = A / \sim_{\text{car}} \cong \Sigma_S(A).$$

Then exactly one identity-stage condition holds:

1. $|Q| = 1$: *standing carrier identity is singleton and no further identity selection is meaningful at the fixed target;*
2. $|Q| > 1$: *the non-singleton standing carrier quotient is same-scope final, and structural pre-lock singleton realization requires trace realization or impossibility realization.*

If the second branch holds and neither realization route is certified, the strongest same-scope endpoint is irreducible structural underdetermination, not a preferred carrier.

Proof. The identity-stage dichotomy is cardinal bivalence for the nonempty quotient. The first branch is immediate. In the second, quotient finality closes further same-scope identity refinement, and the preceding theorem exhausts the remaining structural realization routes. If neither is present, no original-certificate route to singleton remains. $\square$

9 Singleton carrier theorem

Theorem 9.1 (Standing Carrier Singleton Theorem). *Let $A \subseteq \mathcal{P}$ be a nonempty live candidate family for an activated target-complete scope $\mathcal{S}$. If*

$$|\Sigma_{\mathcal{S}}(A)| = 1,$$

then

$$|A / \sim_{\text{car}}| = 1.$$

The target therefore has exactly one standing carrier identity across the entire candidate family, even if A contains multiple source/provenance presentations.

Proof. By the carrier cardinality formula,

$$|A / \sim_{\text{car}}| = |\Sigma_{\mathcal{S}}(A)| = 1.$$

$\square$

Corollary 9.2 (Conditional Role-Occupancy Closure consequence). *Suppose additionally that the terminal target has one fixed carrier role; the singleton carrier residue is pointwise eligible; the unique one-role assignment is jointly compatible; the lawful automorphism action preserves the complete target invariant; and the role-realization certificate is complete. Then the compatible assignment-orbit quotient is singleton and realization is licensed at standing carrier resolution.*

Proof. A one-element role set and one-element live carrier residue admit exactly one bijection. Eligibility and joint compatibility make that bijection a valid assignment. Quotienting a singleton assignment set by any lawful group action remains singleton. Role-realization completeness then permits the Role-Occupancy Closure criterion to be applied. $\square$

Theorem 9.3 (Finality of the irreducible non-singleton quotient). *If $|\Sigma_{\mathcal{S}}(A)| > 1$, then $Q = A / \sim_{\text{car}}$ is a genuinely non-singleton final standing-carrier identity quotient for the fixed activated target. No same-scope redefinition, provenance refinement, canonical representative, or auxiliary classifier can reduce it to one identity. A structural pre-lock singleton can then arise only by realized-carrier trace or impossibility realization; a richer witness or exclusion architecture is an explicitly changed target.*

Proof. Signature-distinct classes are separated by an admitted target witness after licensed continuation. Quotienting them together would violate witness soundness; refining or re-authorizing the quotient beneath fixed tensor content violates quotient finality and no-second-classifier closure. The post-quotient realization theorem then leaves trace realization or impossibility realization as the only same-scope structural singleton routes. $\square$

Theorem 9.4 (Standing Carrier Identity Closure capstone). *Let $A \subseteq \mathcal{P}$ be nonempty in an activated target-complete scope and let $Q = A / \sim_{\text{car}}$. Then the carrier-identity status is exact:*

$$|Q| = |\Sigma_S(A)|.$$

If this cardinality is one, the fixed target has exactly one standing carrier identity across A , with downstream realization still subject to its independent eligibility, compatibility, automorphism, and completeness obligations. If the cardinality exceeds one, Q is final for same-scope carrier identity: no alternative quotient, provenance refinement, or second classifier can collapse it. In that non-singleton branch, a structural pre-lock singleton realized-carrier certification under the original target exists exactly through a realized-carrier trace or an impossibility realization; if neither is certified, the strongest endpoint is irreducible structural underdetermination at carrier resolution.

Proof. The cardinality identity is the signature-image formula. The singleton branch is the Standing Carrier Singleton Theorem together with the explicit Role-Occupancy nonpromotion boundary. In the non-singleton branch, finality follows from signature separation and uniqueness of the standing quotient. The post-quotient exhaustion theorem proves necessity of trace or impossibility, while the definitions of those two routes make either route sufficient to fix exactly one already standing class. If both are absent, no original-certificate structural singleton route remains. $\square$

10 Negative control: when collapse must not occur

The theorem is designed not to manufacture singletonhood. A useful negative control is a target with two response classes that are structurally separated but for which the current certificate contains neither a singleton realized-response trace nor an impossibility theorem excluding one class. In that situation the correct result is deferral, not quotient collapse.

The neutrino matter-ordering prediction-lock analysis provides precisely this pattern [16]. Its current certificate contains an orientation-neutral witness ledger and candidate matter-response classes, but the trace-realization route supplies no singleton realized-response trace and the exclusion-realization route supplies no branch impossibility. The analysis therefore exhausts the same-scope realization routes and returns a fail-closed no-pre-lock-prediction status rather than selecting a branch.

Proposition 10.1 (Negative-control consistency). *The standing carrier identity theorem does not collapse a pair of classes merely because one target contains both. If the classes differ in an already admitted standing continuation signature, they remain distinct in Q_{car} . If neither trace nor impossibility realizes one, the correct same-scope status is non-singleton/underdetermined.*

Proof. Signature difference places the alternatives in distinct elements of the final standing quotient and blocks identity collapse by witness soundness. The post-quotient route-exhaustion theorem then requires trace realization or impossibility realization. If both are absent, no same-scope structural singleton route remains. $\square$

11 Counterexample burden and hostile tests

A counterexample to the main theorem must preserve the fixed target and the completeness of the admitted witness/continuation interface while producing a standing-bearing carrier distinction outside the anchor-tensor-skin and signature quotient structure.

Attack	Required construction	Consequence if successful
Hidden source difference	Two source-distinct presentations with identical complete standing signatures, yet a target-internal reason they remain different standing carriers without a second classifier	Refutes stable-reference specialization or witness completeness
Coarse quotient attack	Two presentations separated by an admitted target witness but identified by $\sim_{\text{car}}$	Refutes the signature definition or soundness
Activation failure	A standing-bearing target consequence omitted from the independently frozen consequence-functional / continuation interface	Application gate fails; no carrier count is licensed
Fourth carrier role	A difference that changes neither target typing nor signature yet carries standing-bearing target work	Refutes anchor–tensor–skin exhaustiveness
Second classifier	A same-scope authorizer that refines a signature fiber while preserving bivalence, stable reference, no repair, and no selector import	Refutes no-second-classifier closure
Post-quotient third realization route	A structural singleton from a genuine non-singleton final quotient that is neither realized trace nor impossibility elimination and uses no empirical comparison, imported selector, or scope change	Refutes post-quotient route exhaustion
Source-label primitive authority	A proof that provenance or primitive inventory bears carrier identity merely by being recorded, without changing any admitted target witness or continuation	Refutes carrier descent / source-difference closure

Remark 11.1 (The most important falsifier). *The clean-room theorem should be rejected for a specific application if the allegedly “hidden” source distinction already changes an independently frozen target witness, target-compatible continuation, failure condition, or standing-bearing structural consequence. In that case the distinction is tensor, not skin, and the carrier quotient must remain non-singleton.*

12 Relation to the supplied structural corpus

The theorem is a specialization and synthesis, not a new primitive layer. Its proof uses the following established structural ideas.

- The kernel paper fixes the four roles Adm, St, Ref, Irr, the AMetric boundary consequences, no same-act repair, standing conservation, and fixed-domain anti-weakening.
- *The Structure of Admissibility* proves bivalence of the admissibility-relevant status quotient, standing as reuse-stable admissibility, uniqueness of the admissible interior, no independent standing carriers, and standing conservation.
- The fixed-domain exhaustion theorem defines stable reference, constructs the canonical continuation-signature quotient, and proves intrinsic detectability.
- *Anchor, Tensor, and Skin* supplies the load-bearing descent to the descriptive carrier: elementary effect exhaustion, finite composite closure, tensor-determined retained trajectories, and no residual same-scope refinement beneath fixed tensor content [6].
- Standing–Admissibility Identity Closure supplies the explicit no-second-classifier bridge and finite route collapse for same-domain divergence.
- The bivalence paper proves a local standing-conservation spine, no second same-scope admissibility invariant, no fourth same-scope role, and constructional-life exhaustion.
- The Standard-Model classification theorem contains a domain-specific added-load dichotomy and uniqueness of the standing equivalence relation [8]. The present paper generalizes only the abstract identity logic, under explicit target-completeness hypotheses, rather than importing the Standard-Model conclusion wholesale.
- Ensemble Embedding Collapse supplies an independent multiplicity precedent [9]: multiplicity does not inherit standing merely by being collected, and any standing-bearing plurality must be explicitly gated or collapse to bookkeeping/equivalence.
- The two neutrino reconstruction papers provide successful fixed-scope examples [13, 14] in which same-witness alternatives collapse into a unique reconstruction object.
- Singleton Governance [15] is used only as support for the impossibility of arbitration-based second authority; no carrier-cardinality conclusion is inferred directly from governance cardinality.
- The matter-ordering prediction-lock theorem supplies the negative-control realization logic: candidate separation is not selection, and a current certificate with neither trace realization nor impossibility realization must remain fail-closed.

13 Scope and nonclaims

Nonclaim 13.1 (No universal collapse of microscopic ontology). *The theorem does not assert that source-distinct ultraviolet theories, fields, or models are literally the same physical theory. It classifies identity relative to one fixed target-complete standing interface. A distinction that becomes standing-bearing under a richer target may reappear as a genuine tensor or anchor difference there.*

Nonclaim 13.2 (No witness selection after the fact). *The theorem cannot be used by choosing a witness family that deliberately forgets an inconvenient source distinction. The carrier-interface activation certificate is an upstream proof burden; failure to discharge it blocks application before any carrier cardinality claim is formed.*

Nonclaim 13.3 (No empirical denial). *Empirical comparison can determine which of several standing-distinct carrier classes is realized. The theorem only distinguishes such post-lock determination from a pre-lock structural derivation from the fixed certificate.*

Nonclaim 13.4 (No replacement for source-first generation). *Source-first candidate generation remains necessary. The theorem acts only after candidate formation, because one cannot know in advance whether a source difference is tensor, anchor, or skin at the final target resolution.*

Nonclaim 13.5 (No automatic Role-Occupancy license). *A singleton standing carrier quotient is necessary for a one-role carrier target but does not replace eligibility, joint compatibility, lawful automorphism, or certificate-completeness burdens. The conditional Role-Occupancy corollary applies only after those obligations are independently discharged.*

14 Conclusion

The clean-room result separates three objects that must not be conflated:

$$\text{source-generated presentation} \longrightarrow \text{standing carrier identity} \longrightarrow \text{realized carrier class}.$$

The first may remain highly non-singleton for completeness and provenance. The second is fixed by the unique complete standing continuation quotient once the activation gate is discharged. If that quotient is singleton, the identity stage is closed. If it is non-singleton, the quotient is final for same-scope identity and the realized-carrier stage requires either a target-admitted trace or impossibility elimination.

The principal theorem can be compressed to the following sequent:

$$\begin{aligned} & \text{fixed target} + \text{carrier-interface activation} + \text{licensed continuations} \\ & + \text{stable reference} + \text{carrier descent} + \text{bivalence} + \text{no repair} + \text{no second classifier} \\ & \implies Q_{\text{car}}(\mathcal{S}) \cong \text{im}(\Sigma_{\mathcal{S}}), \\ & \text{finite source difference} \implies \text{anchor motion} + \text{tensor content} + \text{skin remainder}, \\ & |Q_{\text{car}}| > 1 \implies \text{same-scope identity finality}, \\ & \text{post-quotient structural singleton routes} = \{\text{trace}, \text{impossibility}\}. \end{aligned}$$

Accordingly, the decisive structural question for any later carrier application is not whether two candidates have different source labels. It is first whether the application has independently activated the complete target standing interface and, if so, whether the candidate difference survives in its continuation signature. If not, carrying the difference into terminal carrier identity is over-resolution. If yes, the distinction is genuine and cannot be removed by any same-scope quotient choice; once the final quotient is non-singleton, singletonhood requires trace realization or impossibility exclusion.

This is the exact constraint needed to test source-distinct carrier residues without importing a preference, changing the target silently, or confusing candidate multiplicity with standing carrier multiplicity.

A Dependency map

Imported result family	Role in the present theorem
Non-Degenerate Construction and the Kernel of Admissibility	Fixed-domain roles Adm, St, Ref, Irr; no same-act repair; AMetric/no-selector consequences; standing conservation and anti-weakening.
The Structure of Admissibility	Bivalence; standing as reuse-stable admissibility; no independent standing carrier; unique admissible interior; conservation.
A Fixed-Domain Exhaustion Theorem	Stable reference; full continuation signature; canonical quotient; intrinsic detectability.
Anchor, Tensor, and Skin	Carrier descent; elementary role exhaustiveness; finite composite closure; tensor-determined retained trajectories; no residual same-scope refinement beneath fixed tensor content.
Standing–Admissibility Identity Closure	No second same-domain classifier; finite divergence-route logic.
Bivalence Theorem for Non-Degenerate Construction	No second same-scope admissibility invariant; no fourth role; constructional-life exhaustion.
Classification Theorem for Admissible Standard-Model Structures	Independent precedent for added-load dichotomy and uniqueness of standing identity quotient; used as corroboration, not as a domain-general premise.
Ensemble Embedding Collapse	Independent multiplicity theorem: inherited multiplicity alone carries no standing; surviving multiplicity must be explicitly gated or bookkeeping.
Common Structural Origin paper	Independent fixed-domain bridge precedent: physically relevant differences must alter exhaustive standing-bearing loci; otherwise they are presentation.
Neutrino reconstruction papers	Domain-specific precedent for same-witness rival exhaustion and quotient uniqueness.
Singleton Governance	Supports rejection of arbitration/second-governor routes only; not used to infer carrier singleton.
Neutrino Ordering Prediction-Lock Closure	Negative-control route exhaustion: candidate separation does not realize a branch; trace or exclusion is required.

B Application protocol

A later application to a concrete carrier residue should proceed in the following order.

1. Freeze the target before reading the desired carrier endpoint.
2. Build the carrier-interface activation certificate: target/definedness interface, complete target-admitted standing consequence family, failure/compatibility loci, and licensed continuation monoid.

3. Apply the carrier-interface activation theorem. If the gate fails, stop: no terminal carrier count is licensed.
4. Preserve the full source-first candidate universe and all provenance labels.
5. Decompose composite source differences componentwise and record anchor motion, tensor-bearing content, and skin/provenance remainder.
6. For every pair of live source presentations, compute the complete standing continuation signature.
7. Form the final standing carrier quotient by signature equality, not by source labels and not by phenomenological preference.
8. For every source distinction discarded by the quotient, record why it is target-inert skin/provenance at the fixed scope; for every surviving distinction, identify the exact already-admitted witness/continuation coordinate that separates it.
9. If the quotient is singleton, apply any downstream role-occupancy theorem only after eligibility, joint compatibility, automorphism, and completeness checks are independently discharged.
10. If the final quotient remains non-singleton, identity collapse is closed. Test realized trace and impossibility realization. If neither exists, report irreducible structural underdetermination at the fixed target.
11. Any new witness, exclusion object, or interface coordinate introduced after this calculation is a target extension and must trigger a fresh certificate rather than retroactively editing the original quotient.

C Compact theorem package

Theorem C.1 (SCIC-0: Carrier-interface activation). *A terminal carrier count is licensed only after an independently formed target interface is shown to contain the complete definedness, quotient-factorizing standing, compatibility, failure, and licensed-continuation loci. Under carrier descent and no refinement beneath tensor content, that activation certificate implies target completeness of the standing continuation signature.*

Theorem C.2 (SCIC-1: Unique carrier quotient). *For an activated target-complete fixed carrier scope, equality of complete standing continuation signatures is the unique admissible terminal standing-carrier identity relation.*

Theorem C.3 (SCIC-2: Source-difference closure). *Every elementary source variation is anchor, tensor, or skin. Every finite composite source difference decomposes into anchor motion, tensor-determined same-scope content, and skin/provenance remainder; no fourth carrier effect channel exists.*

Theorem C.4 (SCIC-3: No hidden source/classifier promotion). *Provenance, primitive source inventory, simplicity, representative choice, or another auxiliary datum cannot refine terminal carrier identity beneath a fixed signature fiber. Such a datum either factors through the standing quotient, is skin, is an inadmissible second same-scope classifier, or changes the target/interface.*

Theorem C.5 (SCIC-4: Carrier-distinction adjudication). *Every proposed carrier distinction is either target/scope change, standing-quotient collapse, or witnessed same-scope distinction. Retaining*

a signature-equivalent pair as distinct by auxiliary authority is a failure normal form, not a fourth semantic carrier status.

Theorem C.6 (SCIC-5: Post-quotient realization exhaustion). *Let $Q = A / \sim_{\text{car}}$ be the final standing carrier quotient of an activated fixed target. If $|Q| > 1$, no further same-scope identity collapse is available. A structural pre-lock singleton under the original certificate can then arise only through realized-carrier trace or impossibility elimination. Empirical comparison is post-lock; selector import is inadmissible; scope change yields a new certificate.*

Theorem C.7 (SCIC-6: Singleton criterion and finality). *For a nonempty live candidate family A ,*

$$|A / \sim_{\text{car}}| = |\Sigma_S(A)|.$$

Hence $|\Sigma_S(A)| = 1$ is necessary and sufficient for singleton standing-carrier identity at the fixed target. If $|\Sigma_S(A)| > 1$, the non-singleton quotient is final for same-scope identity, and absent trace or impossibility the strongest structural endpoint is underdetermination.

References

- [1] A. J. Maley, *Non-Degenerate Construction and the Kernel of Admissibility: An Exhaustion Theorem for Standing, Reference, and Irreversibility*, 2026.
- [2] A. J. Maley, *The Structure of Admissibility: Ametricity, Bivalence, and the Unique Admissible Interior*, 2026.
- [3] A. J. Maley, *A Fixed-Domain Exhaustion Theorem: Standing, Admissibility, and the Determination of Consequence under Typed Boundary Discipline*, 2026.
- [4] A. J. Maley, *Standing-Admissibility Identity Closure: A Fixed-Domain Meta-Necessity Closure*, 2026.
- [5] A. J. Maley, *The Bivalence Theorem for Non-Degenerate Construction: Standing Conservation, Governance Necessity, and Local Closure*, 2026.
- [6] A. J. Maley, *Anchor, Tensor, and Skin: A Fixed-Domain Decomposition Theorem for Admissible Description*, 2026.
- [7] A. J. Maley, *The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissibility-Bearing Construction*, 2026.
- [8] A. J. Maley, *A Classification Theorem for Admissible Standard-Model Structures: Uniqueness Under Standing-Preserving and Transport Constraints*, 2026.
- [9] A. J. Maley, *Ensemble Embedding Collapse under Standing-Preserving Admissibility: Transport, Localization, Operator Exhaustion, and the Uniqueness of the Admissible Interior*, 2026.
- [10] A. J. Maley, *Admissibility and the Common Structural Origin of Gauge, Constraint, and Quantum Structure*, 2026.
- [11] A. J. Maley, *Unus Solus Possibilis Est: Corpus Closure and Uniqueness of the Admissible Interior*, 2026.

- [12] A. J. Maley, *Gauge Coupling Parameters and Forced Uniqueness*, 2026.
- [13] A. J. Maley, *Neutrino Oscillation as the Unique Admissible Reconstruction of Flavor Under Noncommuting Propagation*, 2026.
- [14] A. J. Maley, *Neutrino Oscillation as Boundary-Kernel Reconstruction*, 2026.
- [15] A. J. Maley, *The Necessity of Singleton Governance*, 2026.
- [16] A. J. Maley, *Neutrino Ordering Prediction-Lock Closure under Observable Transport: A Synthesis of the AOT2–AOT7 Matter-Envelope Audit*, 2026.